THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KILIMANJARO REGIONAL COMMISSIONER'S OFFICE

FORM FOUR MOCK EXAMINATION

032/2A **CHEMISTRY 2A**
(ACTUAL PRACTICAL)
MARKING SCHEME
(For Both School and Private Candidates)

May 2024

QUESTION 1(06 mark)

Burette readings (cm^3)	Pilot	1	2	3
Final reading (cm^3)	15.40	30.60	45.60	15.00
Initial reading (cm^3)	0.00	15.40	30.60	0.00
Volume used of acid (cm^3)	15.40	15.20	15.00	15.00

Volume of pipette used was **25.00cm^3** (**01 mark)**

Note: an error of ∓ 0.5cm^3 must be considered

(a) Find the average volume of "TT"

Volume used of acid = $\frac{15.20+15.00+15.00}{3}$

$=15.00cm^3$

Volume used of acid (va) = 15.00cm^3 (**01 mark)**

(b) i. 15.00cm^3 of solution TT required 25.00cm^3 of solution RR for complete neutralization **(01marks)**

OR:

12.00cm3 of solution TT required 20.00cm^3 of RR

ii. The colour change at the end point was **from pink to colourless** (**01 mark)**

(c) The best indicator to be used is P.O.P since the reaction involve **strong base** and **weak acid**

(d) With state symbols a balanced

Molecular eqn: $2NaOH(aq) + H_2C_2O_4(aq) \longrightarrow Na_2C_2O_4(aq) + 2H_2O_{(1)}$ (**01 mark)**

Total ionic: $2Na^+, 2OH^- + 2H^+, C_2O_4^{2-} \longrightarrow 2Na^+, C_2O_4^{2-} + 2H_2O$ **(1mark)**

Net ionic: $2OH^-(aq) + 2H^+(aq) \longrightarrow 2H_2O(l)$ **(1 mark**)

(e) Required number of moles of TT reacted:

From, dilution law

McVc = MdVd

Vc= 50cm3 = 0.05dm3

Mc= 1.5M

Vd = 5dm3

Md=

$Md = \frac{McVc}{Vd}$ (**0.5 mark)**

$= \frac{1.5 \times 0.05}{5}$

Md = 0.015M

Molarity of diluted acid (Md) = 0.015M **(01 mark)**

(f) Calculate percentage purity of sodium hydroxide (NaOH)

From

$\frac{MaVa}{MbVb} = \frac{na}{nb}$

$2NaOH_{(aq)} + H_2C_2O_{4\,(aq)} \longrightarrow Na_2C_2O_{4(aq)} + 2H_2O_{(1)}$

na: nb = 1:2 (0.5 mark)

$Mb = \frac{MaVanb}{Vbna}$ **(0.5 mark)**

$Mb = \frac{0.015\ x\ 15x2}{25x1}$

Mb= 0.018M

Molarity of base (NaOH) = 0.018M **(01 mark)**

Calculate concentration of pure (NaOH)

From

Molarity = $\frac{concentration\ (g/dm3)}{molar\ mass\ (NaOH)g/mol}$ **(0.5 mark)**

Conc (g/dm3) = molarity x Mr.

= 0.018mol/dm^3 x (23+16+1)g/mol

= 0.018mol/dm^3 x 40g/mol **(1 mark)**

= 0.72g/dm^3

Concentration of pure (NaOH) = 0.72g/dm^3 **(01 mark)**

Calculate concentration of impure (NaOH)

Mass of impure (NaOH) = 0.45g

Volume of solution = 0.5dm3

Conc = $\frac{mass\ (g)}{volume\ (dm3)}$ **(0.5 mark)**

$= \frac{0.45g}{0.5dm3}$

Conc = 0.9g/dm^3 **(01 mark)**

Concentration of impure (NaOH) = 0.9g/dm3

From

%purity = $\frac{conc.pure}{conc.impure}$ x 100% **(0.5 mark)**

$= \frac{0.72}{0.9}$x 100%

=80% **(01 mark)**

Percentage purity of NaOH = 80% ***note***: by considering an error of $\mp$ 0.5cm3

If Va = 14.50cm3, % purity = 75.5%

If Va = 15.50cm3, % purity = 82.6%

(c) Application of neutralization reaction between TT and RR in daily life;

i. Used to treat insect stings(**01 mark)**
ii. Used to neutralize acidic soil(**01 mark)**
iii. Relieving indigestion
iv. Used to treat factory wastes(**any two 02 mark)**

(d) Source of error

i. Parallax error **(0.5 mark)**
ii. Using contaminated solution (**0.5 mark)**
iii. Amount of indicator used
iv. calibration error due to instrument(instrumental error)
v. Too much approximation
vi. Losing solution due to swirling **(any two 01 mark)**

QUESTION 2

QUALITATIVE ANALYSIS REPORT: (15 marks each row 02 marks)

S/N	EXPERMENT	OBSERVATION	INFERENCE
a.	Sample Q was observed	- White -powdered solid	$NH4^{+}$, Na^{+}, $Ca2^{+}$,Pb^{2+} , Zn^{2+}may be present (**2mark**)
b.	A small amount of solid sample Q was heated in a dry test –tube	- A colourless gas evolved which turned lime water milky and moist litmus paper from blue to red	CO_3^{2-} and HCO^{-}_3 might be present **(02 marks)**
c.	A small amount of solid sample was taken into a dry test tube then distilled water was added and the mixture was shaken	- Insoluble when cold	CO_3^{2-} of Ca^{2+} , Pb^{2+} , $2n^{2}$ +, Fe^{2+} , Fe^{3+} Cu^{2+} might be present or SO_4^{2-} of Ca^{2+} Pb^{2+} might be present **(02 marks)**
d.	A small amount of solid sample Q was taken into a dry test –tube then diluted hydrochloric acid was added until it is half full.	Effervescence of colourless gas evolved which turned lime water milky and moist litmus paper from blue to red	CO_3^{2-} , HCO_3^{-} Might be present **(02 marks)**
e.	The resulting solution in (d) above was divided into two portions i. To the first portion ammonia solution was added dropwise till in excess	- No precipitate formed	Ca^{2+}, Na^{+} might be present **(02 marks)**

	ii. To the second portions sodium hydroxide solution were added dropwise till in excess	White precipitate was formed, insoluble in excess NaOH	Ca^{2+} might be present **(02 marks)**
f.	A small amount of solid sample Q was transferred in dry test- tube then a small volume of dilute nitric acid was added	- Effervescence of colourless gas, which turned lime water milky	CO_3^{2-} was present and confirmed **(02 marks)**
g.	A flame test was performed for solid sample Q	- Brick - red flame	Ca^{2+} was present and confirmed **(02mark)**

Conclusion

a. i. The cation in sample Q was Ca^{2+} (**01 mark)**
   ii. The anion in sample Q was CO_3^{2-} **(01 mark)**
   iii. The chemical formula of sample Q was $CaCO_3$ **(0.5 mark)**
   iv. Sample Q was calcium carbonate **(0.5 mark)**

b. Chemical equation for the reaction took place in (b) above

$$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2(g)}$$ **(02 marks)**

Chemical equations for (d) above

$$CaCO_{3(s)} + 2HCl_{(aq)} \longrightarrow CaCl_{2(aq)} + HO_{2(1)} + CO_{2(g)}$$ **(2marks)**

c. Uses of Q ($CaCO_3$)
   i. Used in manufacture of calcium silicate($CaSiO_3$) which is used in manufacture of glass **(01 mark)**
   ii. Used in manufacture of cement **(01 mark)**
   iii. Used to neutralize treat acidic soil **(any two 02 mark)**